

Google Cloud Run Deployment Guide

Production deployment with Cloud Run, PostgreSQL, Redis, Secret Manager, and GitHub Actions

Purpose

Deploy the Spring Boot URL shortener to Google Cloud Run using a production Docker image, Cloud SQL PostgreSQL, Memorystore Redis, Secret Manager, Artifact Registry, and GitHub Actions on every push to main.

Architecture

- ? Cloud Run hosts the Spring Boot container.
- ? Artifact Registry stores Docker images tagged by Git commit SHA.
- ? Cloud SQL PostgreSQL stores URL and analytics data.
- ? Memorystore Redis stores shortCode to originalUrl cache entries.
- ? Secret Manager stores database username and password.
- ? Serverless VPC Access lets Cloud Run reach Memorystore private IP.
- ? GitHub Actions uses Workload Identity Federation, not a service account key.

Generated Files

File	Purpose
.github/workflows/deploy-cloud-run.yml	Tests, builds, pushes, and deploys on push to main.
deploy/gcloud-setup-template.ps1	Creates APIs, Artifact Registry, Cloud SQL, Redis, VPC connector, service accounts, and secrets.
deploy/workload-identity-setup-template.ps1	Creates GitHub OIDC identity provider.
deploy/github-actions-variables.md	Lists required GitHub variables and secrets.
deploy/cloud-run-service.yaml	Reference Cloud Run service manifest.
.dockerignore	Keeps the production Docker context small.

Step 1 - Prepare Google Cloud

```
gcloud auth login
gcloud auth application-default login
```

Step 2 - Create Google Cloud Resources

Edit deploy/gcloud-setup-template.ps1, replace YOUR_PROJECT_ID and region values, then run it. The database password is entered interactively and stored in Secret Manager.

```
cd C:\Users\dell\Documents\Codex\url-shortener-ai-assignment
.\deploy\gcloud-setup-template.ps1
```

Resource Checklist

Resource	Recommended name
Artifact Registry	url-shortener
Cloud SQL PostgreSQL	url-shortener-postgres
Database	urlshortener
Memorystore Redis	url-shortener-redis
Serverless VPC connector	url-shortener-vpc

Resource	Recommended name
Runtime service account	url-shortener-runtime@PROJECT_ID.iam.gserviceaccount.com
Deploy service account	url-shortener-deploy@PROJECT_ID.iam.gserviceaccount.com

Step 3 - Configure GitHub OIDC

Edit `deploy/workload-identity-setup-template.ps1` with project ID, project number, and OWNER/REPOSITORY, then run:

```
.\deploy\workload-identity-setup-template.ps1
```

Step 4 - Configure GitHub Variables

Variable	Example
GCP_PROJECT_ID	my-gcp-project
GCP_REGION	us-central1
CLOUD_RUN_SERVICE	url-shortener
CLOUD_RUN_RUNTIME_SERVICE_ACCOUNT	url-shortener-runtime@my-gcp-project.iam.gserviceaccount.com
ARTIFACT_REPOSITORY	url-shortener
APP_BASE_URL	https://your-service-url.a.run.app
CLOUD_SQL_CONNECTION_NAME	my-gcp-project:us-central1:url-shortener-postgres
CLOUD_SQL_DATABASE	urlshortener
SERVERLESS_VPC_CONNECTOR	url-shortener-vpc
REDIS_HOST	Memorystore private IP
REDIS_PORT	6379
DB_USERNAME_SECRET	url-shortener-db-username
DB_PASSWORD_SECRET	url-shortener-db-password

GitHub Secrets

Secret	Value
GCP_WORKLOAD_IDENTITY_PROVIDER	Full provider resource name printed by setup script
GCP_DEPLOY_SERVICE_ACCOUNT	url-shortener-deploy@PROJECT_ID.iam.gserviceaccount.com

Step 5 - Deploy Automatically

```
git add .
git commit -m "Add Cloud Run deployment"
git push origin main
```

Step 6 - Verify

```
Invoke-RestMethod https://YOUR_CLOUD_RUN_URL/actuator/health
```

```
Invoke-RestMethod `
-Method Post `
-Uri https://YOUR_CLOUD_RUN_URL/api/v1/urls `
-ContentType application/json `
-Body '{"originalUrl":"https://www.google.com"}'
```

```
curl.exe -i https://YOUR_CLOUD_RUN_URL/YOUR_SHORT_CODE
```

```
Invoke-RestMethod https://YOUR_CLOUD_RUN_URL/api/v1/analytics/YOUR_SHORT_CODE
```

Security Checklist

- ? Never commit passwords, service account keys, or generated credential files.
- ? Keep database credentials in Secret Manager.
- ? Use Workload Identity Federation for GitHub Actions.
- ? Grant least privilege to runtime and deploy service accounts.
- ? Review `--allow-unauthenticated` before production release.
- ? Use a custom domain and update `APP_BASE_URL`.